

Particle in a hyperbolic space with magnetic field

Ruihua Fan

Abstract

In this note, we study how a charged particle moves in a hyperbolic space and what happens when there is a constant magnetic field.

1 Particle in a Hyperbolic disk

The hyperbolic disk is a unit disk on the complex plan endowed with the metric

$$ds^2 = \frac{4dzd\bar{z}}{(1 - z\bar{z})^2}, \quad |z| < 1. \quad (1)$$

where $z = x + iy$ is the complex coordinate. We would like to find the geodesics between two points and its length. Physically, this is study a particle in an Euclidean AdS_2 space. It is always possible to solve the Euler-Lagrange equation directly. Here, we will find the answer by exploiting several properties of the hyperbolic disk.

The hyperbolic disk can be mapped to the upper half-plane by the following Mobius transformation

$$w = i \frac{1 - z}{1 + z}, \quad z = \frac{i - w}{i + w}, \quad (2)$$

where w denotes the coordinate for the UHP. Accordingly, the metric becomes¹

$$ds^2 = \frac{dw d\bar{w}}{(\text{Im } w)^2} = \frac{dx^2 + dy^2}{y^2}. \quad (3)$$

which is called as hyperbolic/Poincaré plane. The mapping between is depicted in Fig. 1

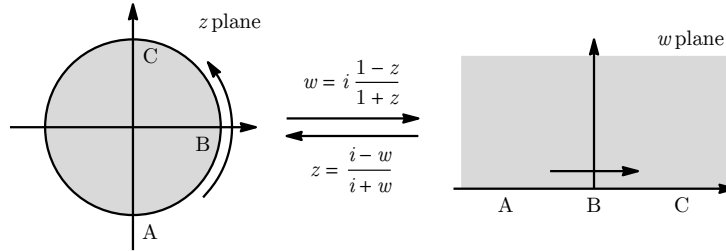


Figure 1: The conformal mapping between the unit disk and the upper half plane.

Solving the geodesic on this UHP turns out to be much easier. The action and the corresponding E-L equation are

$$S = \int \sqrt{\frac{dx^2 + dy^2}{y^2}} \Rightarrow \frac{d}{dy} \left(\frac{dx/dy}{y\sqrt{1 + (dx/dy)^2}} \right) = 0, \quad (4)$$

where we consider y as the “time” to utilize the translation symmetry in the x -direction. The solution is a semi-circle centered at $(x_c, 0)$ on the real axis with an arbitrary radius and can be written as

$$(x - x_c)^2 + y^2 = r^2. \quad (5)$$

¹ $dz = \frac{-2idw}{(w+i)^2}$, $d\bar{z} = \frac{-2id\bar{w}}{(\bar{w}-i)^2}$, $1 - z\bar{z} = \frac{2i(w-\bar{w})}{(w+i)(\bar{w}-i)}$. Notice that $w - \bar{w} = 2i \text{Im } w$ and the result follows.

Mapping that back to the hyperbolic disk yields the geodesic on that geometry and we can write the trajectory as

$$\frac{i - (x_c + r \cos \theta + ir \sin \theta)}{i + (x_c + r \cos \theta + ir \sin \theta)}. \quad (6)$$

Noticing that the transformation is conformal and thus preserves the angle, the trajectory will be some smooth curve which goes perpendicular to the unit circle near the two ends. Some typical examples are shown in Fig. 2

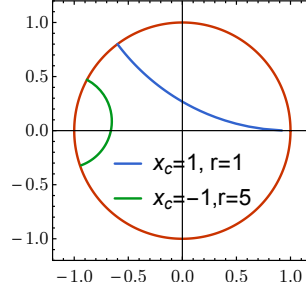


Figure 2

Now, we compute the length of the geodesic between two arbitrary points z_1 and z_2 . Before that, we take a detour to show the following property: the metric is invariant under any conformal automorphism of the unit disk. This is because any $\text{Aut}(\mathbb{D})$ can be written as the following Möbius transformation

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \bar{\alpha}z}, \quad |\alpha| < 1, \quad (7)$$

which basically interchanges $z = 0$ and $z = \alpha$ and its global effect is shown in Fig. 3. One can easily show that ds^2 is invariant under this map.

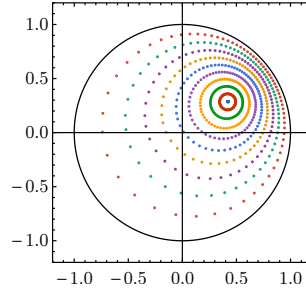


Figure 3: How the concentric circles are mapped under the automorphism. One can see that $z = 0$ is mapped to α followed by some rotation. The concentric circles are deformed accordingly.

Given any two points z_1, z_2 , we are always able to pick one $\text{Aut}(\mathbb{D})$ to map z_1 to the origin and z_2 to some point x_{12} on the positive real axis. Due to the rotation symmetry, the distance between the origin and $x_{12} \in [0, 1)$ is easy to calculate

$$d(0, s) = \int_0^{x_{12}} dx \frac{2dx}{1-x^2} = \log \frac{1+x_{12}}{1-x_{12}}. \quad (8)$$

Now, we only have to find the $\text{Aut}(\mathbb{D})$ that does the job for us. Mapping z_1 to the origin fixes $\alpha = z_1$ and so that we have

$$x_{12} = \text{Abs } f(z_2) = \text{Abs } \frac{z_1 - z_2}{1 - \bar{z}_1 z_2}. \quad (9)$$

To be concrete, let us look at an extreme limit, where both z_1 and z_2 are close to the unit circle. Without loss of generality we choose $z_1 = 1 - \epsilon$, $z_2 = (1 - \epsilon)e^{i\phi}$ with $\epsilon \ll 1$. Then we have

$$x_{12} = \frac{(1 - \epsilon)2 \sin \frac{\phi}{2}}{\sqrt{1 + (1 - \epsilon)^4 - 2(1 - \epsilon)^2 \cos \phi}} \approx \frac{1 - \epsilon}{\sqrt{1 - \epsilon + \frac{\epsilon^2}{2 \sin^2 \frac{\phi}{2}}}} \approx 1 - \frac{\epsilon^2}{2 \sin^2 \frac{\phi}{2}}. \quad (10)$$

and the geodesic distance

$$d(\phi) \approx 2 \log \left(\frac{2}{\epsilon} \sin \frac{\phi}{2} + \mathcal{O}(1) \right). \quad (11)$$

It already has the flavor of the entanglement of the ground state of a CFT on a circle.

One can use this result to infer the semi-classical limit of a boundary-boundary two point function for a massive scalar field, which should be

$$G(\phi_1, \phi_2) \sim e^{-md(\phi_1, \phi_2)} \sim \left(\sin \frac{\phi_{12}}{2} \right)^{-2m} \sim \phi_{12}^{-2m} \Rightarrow \Delta = 2m. \quad (12)$$

2 Charged particle in the Hyperbolic disk with a constant magnetic field

We extend the previous problem by adding a constant magnetic field. As we have seen before, without magnetic field the geodesic always bends and dives towards the boundary $\partial\mathbb{D}$ perpendicularly. The Lorentz force also has a similar effect of bending the trajectory. Therefore, we would expect a competing between the magnetic field and the background curvature. Our discussion mostly follows [this paper](#).

The Euclidean action for a relativistic particle coupled to an external gauge field is

$$S = \int ds + q \int d\tau A_\mu \frac{dx^\mu}{d\tau}, \quad (13)$$

where ds is the proper length and q denotes the charge. However, ds involves a square root and makes the action hard to tackle. Noticing that we finally impose $\sqrt{\dot{x}_\mu \dot{x}^\mu} = 1$, it is mathematically equivalent to use a quadratic action

$$S = \int d\tau \left(\frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu + q A_\mu \dot{x}^\mu \right). \quad (14)$$

One can show that the geodesic equation becomes

$$\frac{d}{d\tau} \left(g_{\mu\sigma} \frac{dx^\mu}{d\tau} \right) - \frac{1}{2} \partial_\sigma g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = q F_{\sigma\mu} \frac{dx^\mu}{d\tau}, \quad F_{\sigma\mu} = \partial_\sigma A_\mu - \partial_\mu A_\sigma, \quad (15)$$

in which the Lorentz force appears on the right hand side. In a curved spacetime, it is more natural to define a constant magnetic field by the flux, namely the magnetic flux per unit area is a constant. In our 2D case, we have

$$F = B \sqrt{g} dx^1 \wedge dx^2 \quad (16)$$

with B being a constant, which is also by construction covariant.

With these preliminaries, we solve the problem on the Poincaré plane and map the solution back to the unit disk. On the UHP, we choose the Landau gauge so that the vector potential and magnetic field are

$$A = A_x dx = \frac{B}{y} dx \Rightarrow F = \frac{B}{y^2} dx \wedge dy \quad (17)$$

The Lagrangian takes the following form

$$L = \frac{\dot{x}^2 + \dot{y}^2}{2y^2} + \frac{qB}{y} \dot{x}. \quad (18)$$

A quadratic Lagrangian can always be solved by brute force. Before doing that, let us first determine the symmetry of the problem. Without the magnetic field, the system has a $\text{SL}(2, \mathbb{R})$ symmetry (we use w for UHP and z for the unit disk)

$$w \rightarrow \frac{aw + b}{cw + d}, \quad ad - bc = 1, \quad a, b, c, d \in \mathbb{R}. \quad (19)$$

Such an action induces a transformation on the gauge potential. It is obvious that the dilation and translation in x -direction leaves A_μ invariant and we only have to check the effect of special conformal transformation

$$w \rightarrow \frac{w}{\lambda w + 1}, \quad \lambda \in \mathbb{R} \Rightarrow \begin{cases} x \rightarrow \frac{x + \lambda(x^2 + y^2)}{(\lambda x + 1)^2 + \lambda^2 y^2} \\ y \rightarrow \frac{y}{(\lambda x + 1)^2 + \lambda^2 y^2} \end{cases}. \quad (20)$$

Accordingly, the gauge potential picks up gauge transformation

$$A \rightarrow A + d\varphi, \quad \varphi = 2 \arctan \frac{\lambda y}{\lambda x + 1}. \quad (21)$$

This implies that the Lagrangian is changed by a total derivative and therefore the system *still* has a $\text{SL}(2, \mathbb{R})$ symmetry.

It also suffices to look at the infinitesimal transformation

$$\begin{aligned} \text{Dilation: } \delta x &= \epsilon x, \quad \delta y = \epsilon y; \\ \text{Translation: } \delta x &= \epsilon, \quad \delta y = 0; \\ \text{SCT: } \delta x &= -\epsilon(x^2 - y^2), \quad \delta y = -2\epsilon xy, \end{aligned} \quad (22)$$

and verify the Lagrangian is invariant under these infinitesimal transformations up to a total derivative. More instructively, this calculation tells us the corresponding conserved charges and one can find that the charge for SCT is extended by a term related to the magnetic field

$$\begin{aligned} \text{Dilation: } \delta S &= \int d\tau \left(\frac{\partial L}{\partial \dot{x}} x + \frac{\partial L}{\partial \dot{y}} y \right) \dot{\epsilon} \Rightarrow Q_D = xp_x + yp_y; \\ \text{Translation: } \delta S &= \int d\tau \left(\frac{\partial L}{\partial \dot{x}} \right) \dot{\epsilon} \Rightarrow Q_T = p_x; \\ \text{SCT: } \delta S &= \int d\tau \left(-\frac{\partial L}{\partial \dot{x}} (x^2 - y^2) - \frac{\partial L}{\partial \dot{y}} 2xy \right) \dot{\epsilon} + 2qB\dot{y}\epsilon \\ &\Rightarrow Q_S = (x^2 - y^2)p_x + 2xyp_y + 2qBy, \end{aligned} \quad (23)$$

where we have introduced the canonical momentum

$$p_x = \frac{\partial L}{\partial \dot{x}} = \frac{\dot{x}}{y^2} + \frac{qB}{y}, \quad p_y = \frac{\partial L}{\partial \dot{y}} = \frac{\dot{y}}{y^2}. \quad (24)$$

If we introduce the canonical Poisson bracket $\{q_i, p_j\} = \delta_{ij}$, we can verify that the conserved charges satisfy the $\text{SL}(2, \mathbb{R})$ algebra as expected

$$\{Q_D, Q_T\} = Q_T, \quad \{Q_D, Q_S\} = -Q_S, \quad \{Q_T, Q_S\} = -2Q_D. \quad (25)$$

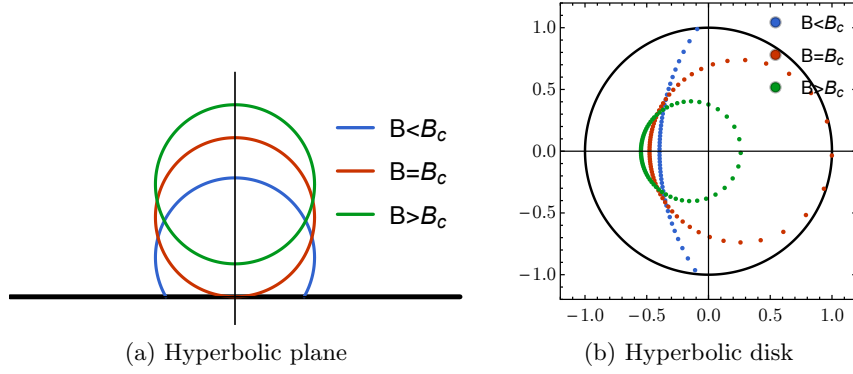


Figure 4: Geodesics on the hyperbolic space with a constant magnetic field.

The Hamiltonian, following the definition, is the quadratic Casimir operator (The conservation of charges and the algebra between them almost implies this, but we still need some calculation to fix the additive constant)

$$H = y^2 \frac{p_x^2 + p_y^2}{2} - qByp_x + (qB)^2 = \frac{Q_D^2 - Q_S Q_T + (qB)^2}{2}, \quad (26)$$

in which the quadratic Casimir can be recognized

$$C = -Q_D^2 + Q_S Q_T. \quad (27)$$

By eliminating p_x, p_y with Q_D, Q_T , the trajectory can be written as

$$\left(x - \frac{Q_D}{Q_T}\right)^2 + \left(y - \frac{qB}{Q_T}\right)^2 = \frac{2H}{Q_T^2} = \frac{-C + (qB)^2}{Q_T^2}. \quad (28)$$

Let us assume $Q_T > 0$. The shape falls into the following three categories:

1. $qB > \sqrt{2H}$, i.e. $C > 0$, the magnetic field is strong enough to trap the particle so that the geodesic is a closed period orbit.
2. $qB < \sqrt{2H}$, i.e. $C < 0$, the magnetic field is weaker than the background curvature so that the geodesic will bend and dive towards the boundary.
3. $qB = \sqrt{2H}$, i.e. $C = 0$, this is the critical case.

One can map the solution back to the unit disk and get the trajectory on the unit disk. Some typical examples are shown in Fig. 4

Now, we look at how the particle moves as a function of time, which can be solved more conveniently using the following Poisson brackets

$$\begin{aligned} \dot{x} &= \{x, H\} = -\frac{1}{2}Q_S + Q_D x - \frac{1}{2}Q_T(x^2 - y^2) \\ \dot{y} &= \{y, H\} = Q_D y - Q_T xy \end{aligned} \quad (29)$$

The discussion on the $SL(2)$ structure inspires us introduce the complex coordinate $w = x + iy$, which satisfies

$$\dot{w} = -\frac{1}{2}Q_S + Q_D w - \frac{1}{2}Q_T w^2 = -\frac{Q_T}{2} \left(w^2 - \frac{2Q_D}{Q_T} w + \frac{Q_S}{Q_T} \right). \quad (30)$$

The solution highly depends on the discriminant of the right hand side

$$\Delta = 4 \left(\frac{Q_D^2}{Q_T^2} - \frac{Q_S}{Q_T} \right) = -\frac{4C}{Q_T^2}, \quad (31)$$

which is also proportional to the Casimir. Therefore, we have the following two scenarios (assuming $Q_T > 0$)

1. $\Delta = 0$ or $C = 0$, the right hand side has a double zero at $w_1 = Q_D/Q_T$ and the solution can be written as

$$\frac{1}{w - w_1} - \frac{1}{w_0 - w_1} = \frac{Q_T}{2} \tau, \quad (32)$$

where w_0 is the initial position at $t = 0$. In the long time limit, we must have $w \rightarrow w_1$ to match the right hand side. It implies that w_1 is a stable fixed point, which sits on the real axis. We can also represent the trajectory on $\mathbb{D}$, which takes the following form

$$\frac{1}{z - z_1} - \frac{1}{z_0 - z_1} = -i \frac{(Q_T - iQ_D)^2}{Q_T} \tau. \quad (33)$$

2. $\Delta \neq 0$ or $C \neq 0$, the right hand side has two distinct zeros at $w_{1,2} = (Q_D \pm \sqrt{-C})/Q_T$ and the solution can be written as

$$\frac{w - w_1}{w - w_2} = e^{-\sqrt{-C}\tau} \frac{w_0 - w_1}{w_0 - w_2}, \quad (34)$$

where w_0 is the initial position at $t = 0$. It follows immediately from the expression that w_1 and w_2 are two fixed points. In the hyperbolic case $C < 0$, w_1 is a stable fixed point and w_2 is unstable, both of which are on the real axis. In the elliptic case $C > 0$, both are on the upper half-plane. The trajectory represented on $\mathbb{D}$ is

$$\frac{z - z_1}{z - z_2} = e^{-\sqrt{-C}\tau} \frac{z_0 - z_1}{z_0 - z_2}. \quad (35)$$